Department of Physics, Shiv Nadar University

PHY-101: Introduction to Physics-1

Monsoon 2018, Mid-Term Exam

Time: 1h 30 min **Marks: 30**

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1. Select an appropriate choice to answer the following questions. Justify your choice with reasoning. No marks will be awarded without reasoning. 3 x 4 = 12

(A) If two sides of a triangle are denoted by two vectors $\vec{A}$ and $\vec{B}$, then the area of the triangle would be,

(i) $\vec{A}.\vec{B}$ (ii) $|\vec{A} \times \vec{B}|$ (iii) $\frac{1}{2}\vec{A}.\vec{B}$ (iv) $\frac{1}{2}|\vec{A} \times \vec{B}|$

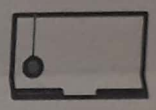
(B) A block is kept on a rough inclined plane. When the incline angle is 30°, the block is about to move. If the angle is 45°, the block starts moving with an acceleration a. If we assume that the value of coefficient of static and kinetic friction are same, the value of a is,

(i) 392 cm/s^2 (ii) 293 cm/s^2 (iii) 392 m/s^2 (iv) 293 m/s^2

(C) A massless spring with spring constant k is attached at one end of a block of mass M that is at rest on a frictionless horizontal table. The other end of the spring is fixed to a wall. A bullet of mass m is fired with a speed v_b into the block from the left and comes to rest in the block. The block and bullet are then immediately start moving together with a speed v. The amplitude of resulting simple harmonic motion is,

(i) $\sqrt{\frac{m+M}{K}}v$ (ii) $\sqrt{\frac{m+M}{K}}v_b$ (iii) $\sqrt{\frac{m+M}{K}}(v+v_b)$ (iv) $\sqrt{\frac{m+M}{K}}(v-v_b)$

(D) A cart of mass M is at rest on a frictionless horizontal surface and a pendulum bob of mass m hangs from the roof of the cart (see figure). The string breaks and the bob falls on the floor and rolls down in a small slot at the middle of the floor. The horizontal distance between the string and the slot is L. The displacement of the cart during this process is,



(i) $\frac{ML}{M+m}$ towards right (ii) $\frac{ML}{M+m}$ towards left (iii) $\frac{mL}{M+m}$ towards right (iv) $\frac{mL}{M+m}$ toward left

2. An object of mass $0.2\ kg$ is hung from a spring whose spring constant is $80\ N/m$. The body is subject to a resistive force given by $-bv$, where v is its velocity (m/sec) and $b = 4\ N\,m^{-1}$ sec.

(a) Set up the differential equation of motion for the oscillations of the system, and find the period of such oscillations for underdamped condition.

(b) Now, the object is subjected to a sinusoidal force given by $F(t) = F_0 \sin\omega t$, where $F_0 = 2\ N$ and, $\omega = 30\ sec^{-1}$. Here, the symbols have their usual meanings. In the steady state, what is the amplitude of the forced oscillation? 5 + 3

3. Two masses (m_1 = 2 kg, m_2 = 5 kg) are moving in same direction with speed v_1 = 10 m/s and v_2 = 3 m/s, respectively. A spring (k = 1120 N/m) is attached at the back of mass m_2. After the collision, the masses are stick together with the spring in between them. Calculate the compression of the spring. Note that initially, m_2 was moving in front of m_1. 5

4. A single conservative force acting on a particle varies as, $\vec{F} = (-Ax + Bx^2)\hat{\imath}$, where A and B are constants and x is in meters. (a) Calculate the potential energy function $U(x)$ associated with this force, taking $U = 0$ at $x = 0$. (b) Find the change in potential energy and the change in kinetic energy as the particle moves from x = 2.0 m to x = 3.00 m. 5

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